

A Replication Study of "Simplicity Bias in 1-Hidden Layer Neural Networks"

Navraj Singh

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1 Introduction

This report presents a comprehensive replication study of the paper "Simplicity Bias in 1-Hidden Layer Neural Networks." The primary objective is to experimentally verify the paper's findings, document any novel observations, and discuss potential reasons for any observed variances from the original results.

The experimental investigation is structured as follows:

1. The claims of simplicity bias are tested on two distinct datasets: a synthetic Independent Features Model (IFM) dataset; a modified MNIST dataset, which is engineered to induce a linearly separable feature set. Further details on the MNIST modification will follow.
2. Models using both "rich" and "lazy" initialization schemes were trained on all three datasets. The number of hidden units in the network was varied to analyze the dynamics of simplicity bias with respect to network width and to approximate the infinite-width neural network limit.
3. The findings from this empirical verification are detailed for all trained models in Section 3.

The complete source code for this project is available on [GitHub](#).

2 Datasets

2.1 Synthetic IFM Dataset

This synthetic dataset was utilized to create a controlled environment with features of increasing complexity. By design, the dataset is linearly separable along the first feature, while the decision boundaries for subsequent features become progressively more non-linear.

2.2 Modified MNIST Dataset

This dataset was created by selecting two classes from the original MNIST dataset and modifying the images to induce linear separability in a specific set of features.

The modification and the expected outcomes are as follows:

1. Images corresponding to class 0 were modified by introducing a patch of white pixels (grayscale value 255) at a fixed location (the first row, comprising 28 pixels). Similarly, a patch of black pixels (grayscale value 0) was introduced in the images for class 1. This modification aims to create a simple, linearly separable feature corresponding to the pixels in the modified patch.
2. We hypothesize that a trained model will primarily rely on the pixels in this modified patch to make its predictions, thereby demonstrating the principle of simplicity bias.
3. This hypothesis will be tested by analyzing the right singular vectors of the first layer's weight matrix to gauge the importance assigned to the patch pixels for classification.

Figure 1 illustrates the modification applied to randomly selected images from the MNIST dataset.



Figure 1: Modified MNIST Dataset. Class 0 images have a white row of pixels at the top, whereas Class 1 images have a black row of pixels.

3 Main Results

The results for each dataset are presented below.

3.1 Synthetic IFM Dataset

For the IFM dataset, our findings show that both rich regime and lazy regime models exhibit simplicity bias.

3.1.1 Rich Regime

The evolution of the effective rank of the first-layer weight matrix is shown in Figure 2 for various numbers of hidden units. It is evident that increasing the number of hidden units leads to a slower convergence of the rank. Key accuracy metrics are provided in Table 1.

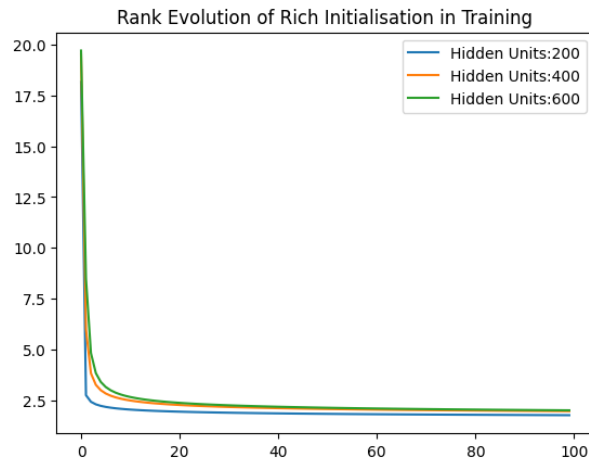


Figure 2: Effective rank evolution for models trained in the rich regime.

The composition of the first two Right Singular Vectors (RSVs) of the first layer’s weight matrix is shown in Figure 3. As the effective rank of the final weight matrix was 2, only the first two vectors were used to construct the projection matrix P .

Hidden Units	rank(P)	Acc(f)	P(orth)-RA	P-RA	P(orth)-LC	P-LC
200	2	99.89	97.53	50.51	29.89	95.16
400	2	99.95	98.66	49.71	35.11	94.37
600	2	100.00	98.97	48.98	38.27	90.76

Table 1: Accuracy metrics for models trained in the rich regime on the IFM dataset.

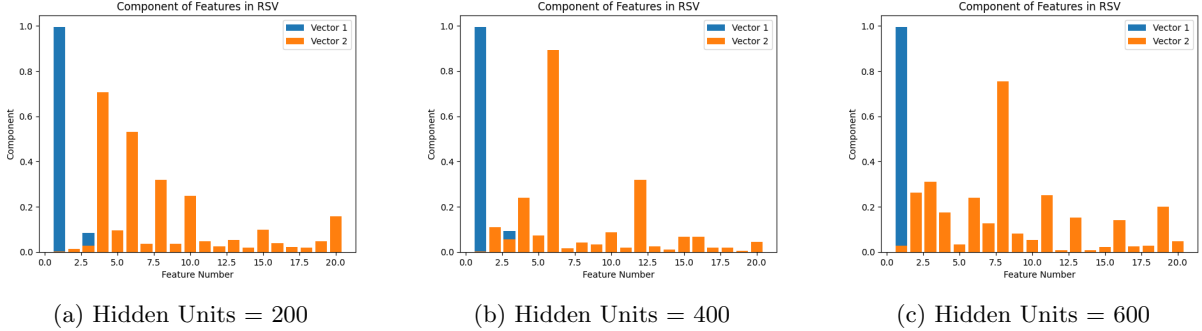


Figure 3: A summary of the first two Right Singular Vectors of the first layer’s weight matrix. The x-axis shows the feature number and the corresponding y-axis gives the component’s absolute magnitude.

3.1.2 Lazy Regime

The training dynamics for the lazy regime are presented in the same format as for the rich regime (Figure 4).

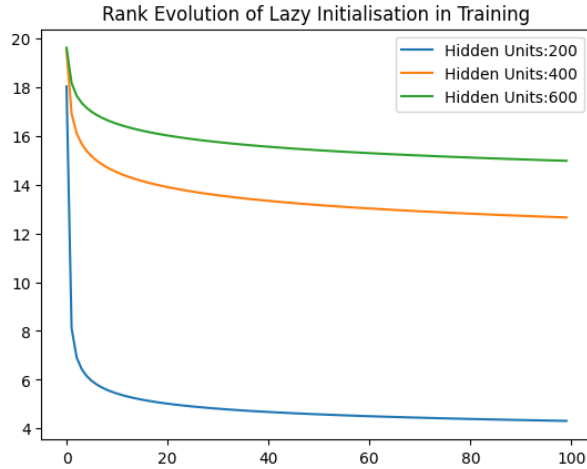


Figure 4: Effective rank evolution for models trained in the lazy regime.

It is evident that the phenomenon of Low-Dimensional Simplicity Bias (LD-SB) truly manifests itself and approaches the theoretical predictions of the infinite-width neural network limit as the number of hidden units increases. The neural network with the highest number of hidden units is most effective at retaining the high rank of its weight matrix, which is characteristic of lazy training.

3.2 Modified MNIST Dataset

As discussed, this dataset was modified to induce linear separability in a set of features, in this case, the first 28 features (corresponding to the flattened first row of pixels). We hypothesize that after training, the first right singular vector will have dominant components corresponding to the first 28 features, and its singular value will be considerably larger than subsequent ones, thus rendering the projection matrix P low-rank.

3.2.1 Rich Regime

In the rich regime, we trained three models with 200, 400, and 600 hidden units on the modified dataset. As hypothesized, the first singular value was larger than the subsequent values by an order of magnitude. Furthermore, the first 28 components of the first RSV, which correspond to the modified pixels, made up more than 60% of the Frobenius norm of the 784-dimensional vector for all models. This strengthens our hypothesis that the models rely on the linearly separable data.

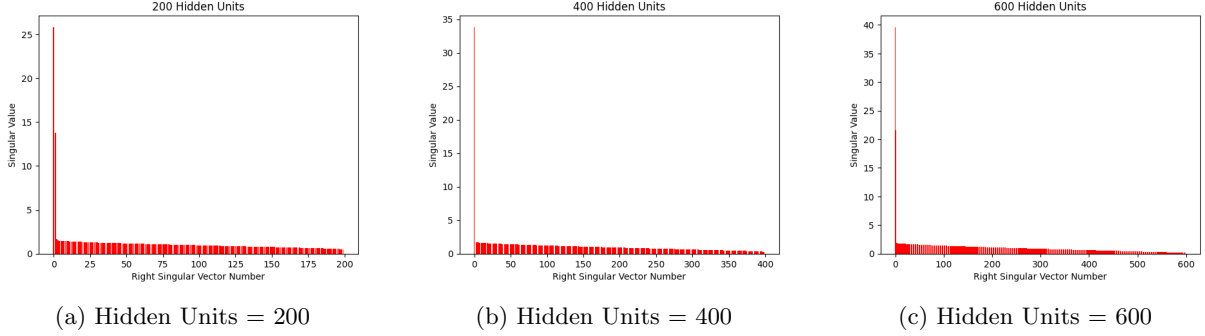


Figure 5: A summary of singular values of the first layer’s weight matrix. It can be observed that the first Right Singular Vector has a dominant singular value. It will be apparent in Figure 6 that this vector is dominated by the first 28 features, proving our hypothesis.

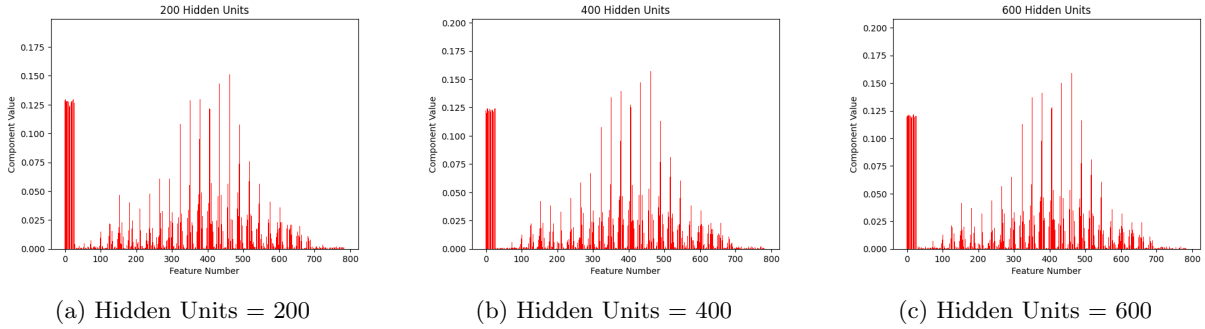


Figure 6: A summary of the first Right Singular Vector of the first layer’s weight matrix. The x-axis shows the feature number, and the corresponding y-axis gives the component’s absolute magnitude.

3.2.2 Lazy Regime

We observed a very interesting and rather contradictory phenomenon while training the lazy models. The **effective rank of the first layer’s weight matrix converged to a very low value of around 5, as evident in Figure 7, which is contradictory to the intrinsic property of lazy training where the rank is expected to remain high.** It is hypothesized that the powerful and consistent gradient signal from the 28 modified pixels forces the weights to change significantly beyond their initialization range, causing the network to enter a feature-learning regime despite the ”lazy” setup.

It should be noted that decreasing the learning rate to values as low as 10^{-6} from the used rate of 10^{-3} had no effect on the rank of the weight matrix; a convergence to a low rank was still observed. The reason a similarly strong gradient signal was not observed in the synthetic IFM dataset is not yet clear and requires further investigation.

Despite this deviation from lazy training dynamics, the characteristic properties of simplicity bias—such as the concentration of the Frobenius Norm in the simple features (70% in this scenario) and the first singular values being very high compared to others—are still observed, as evident in Figures 8 and 9.

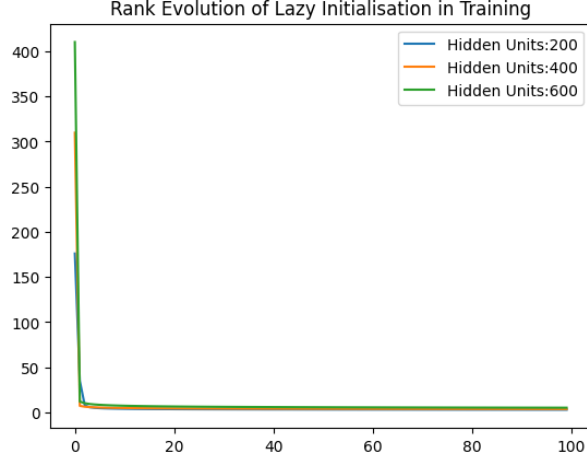


Figure 7: Effective rank evolution for models trained in the lazy regime on the Modified MNIST Dataset.

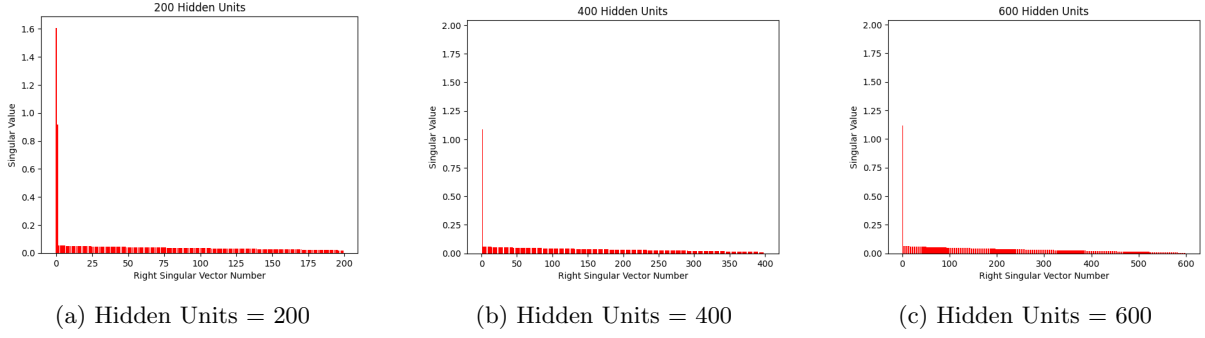


Figure 8: A summary of singular values of the first layer's weight matrix. It is observed that the first Right Singular Vector has a high singular value, which is dominated by the first 28 features (Figure 9), proving our hypothesis.

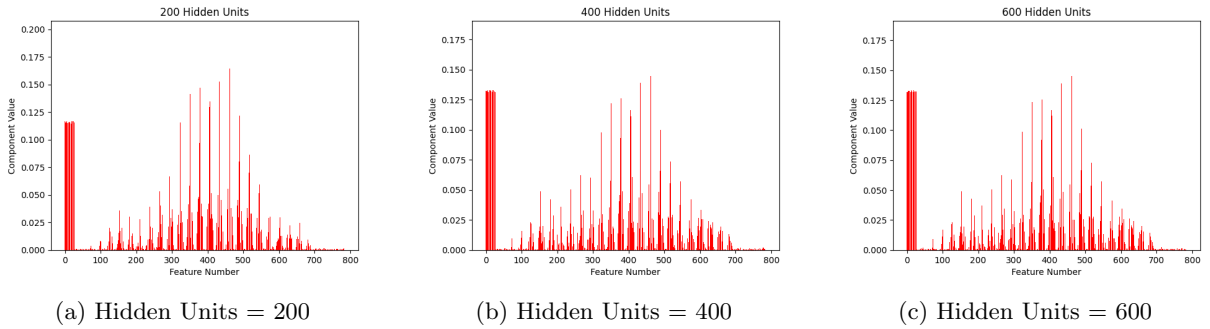


Figure 9: A summary of the first Right Singular Vector of the first layer's weight matrix. The x-axis shows the feature number, and the corresponding y-axis gives the component's absolute magnitude.